
FCN & U-Net

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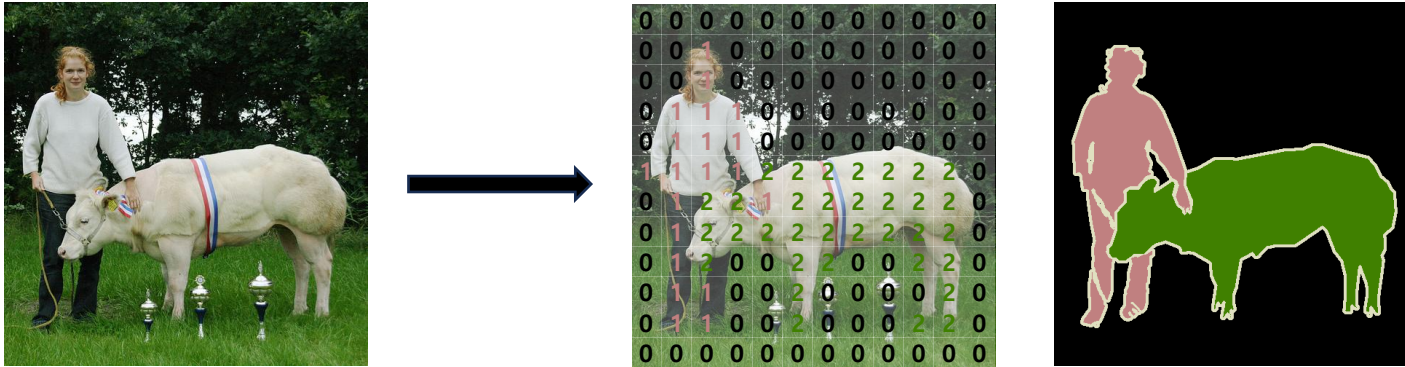
실험 결과

1. Semantic Segmentation
2. FCN
3. U-Net
4. 실험 결과

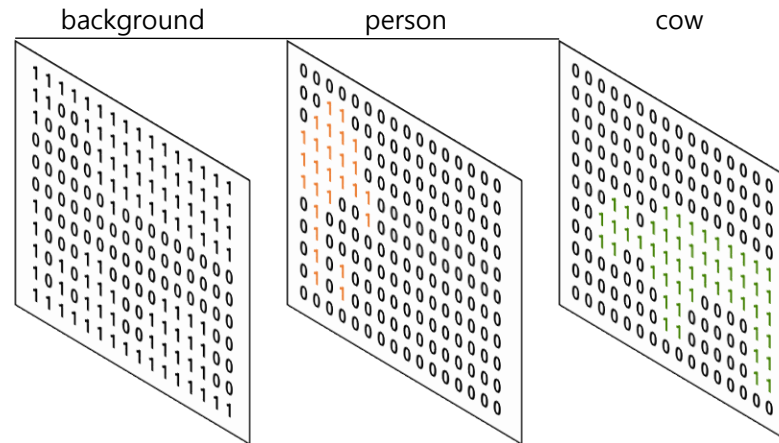
Introduction

■ Semantic Segmentation

- 이미지 내의 픽셀 단위로 Class 예측
- CNN의 recognition에서 localization를 추가로 수행



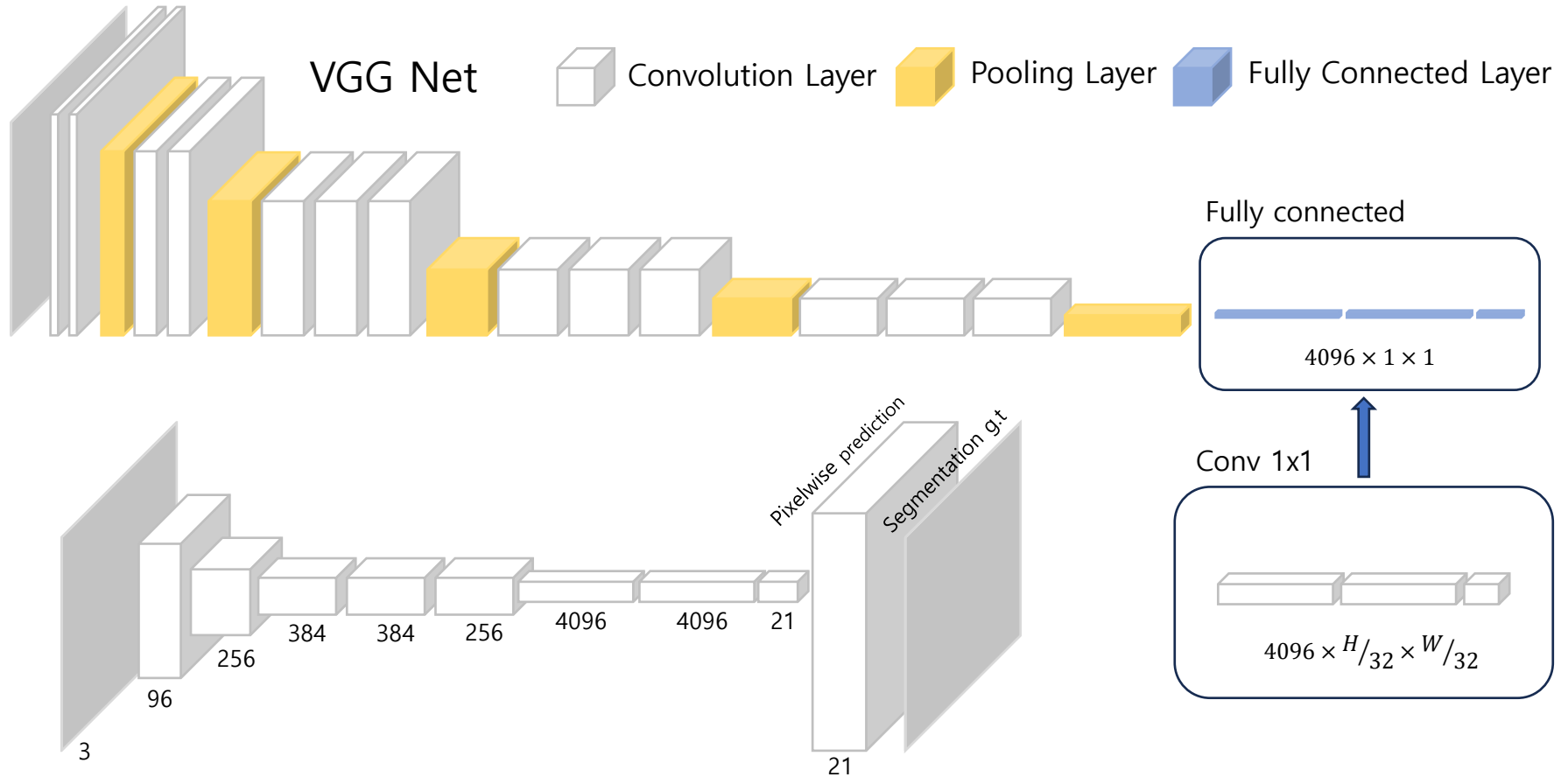
One hot label



FCN^[1]

▪ Main idea

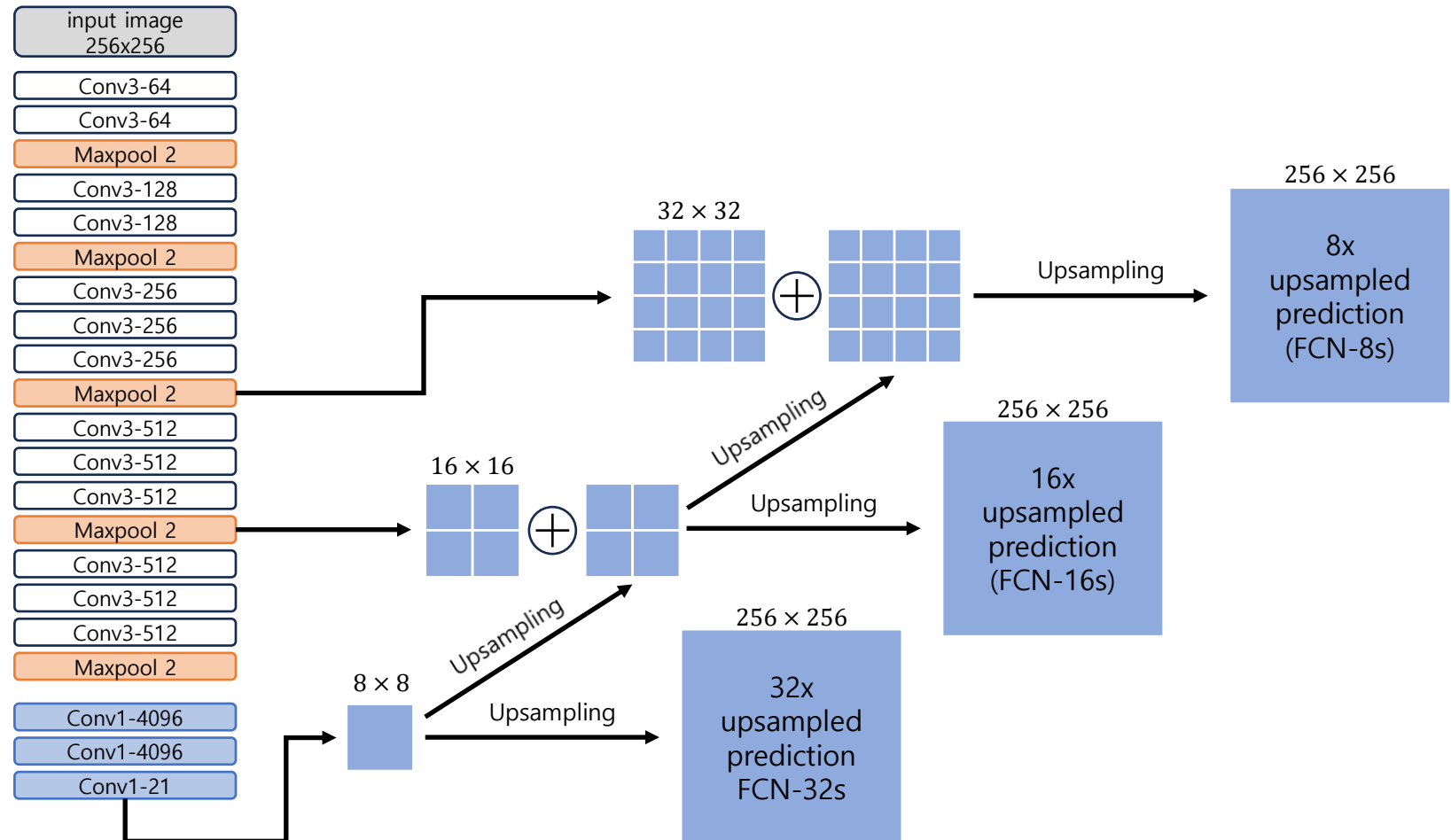
- Classification network를 변형하여 사용
- Location information을 활용하기 위해 FC Layer를 Convolution Layer로 변경



[1] J. Long, E. Shelhamer, T. Darrell. "Fully convolutional networks for semantic segmentation". In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 3431-3440). 2015

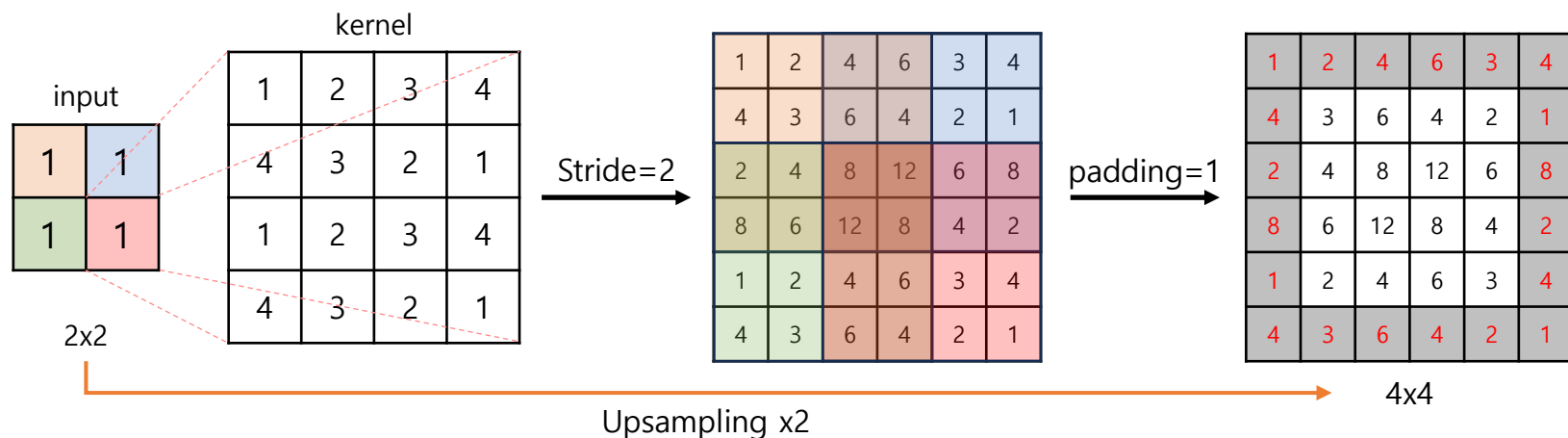
Architecture

- Deep layer에서 얻은 semantic information과 shallow layer에서 얻은 appearance information을 결합



▪ Upsampling

- Deconvolution으로 학습 가능한 파라미터로 취급 가능
- Transposed convolution으로 upsampling에서 학습 가능



$$H_{out} = (H_{in} - 1) \times stride - 2 \times padding + dilation \times (kernel\ size - 1) + output\ padding + 1$$

$stride = 32, padding = 16, dilation = 1, kernel\ size = 64, output\ padding = 0$ FCN-32s : 8x8 $\rightarrow$ 256x256

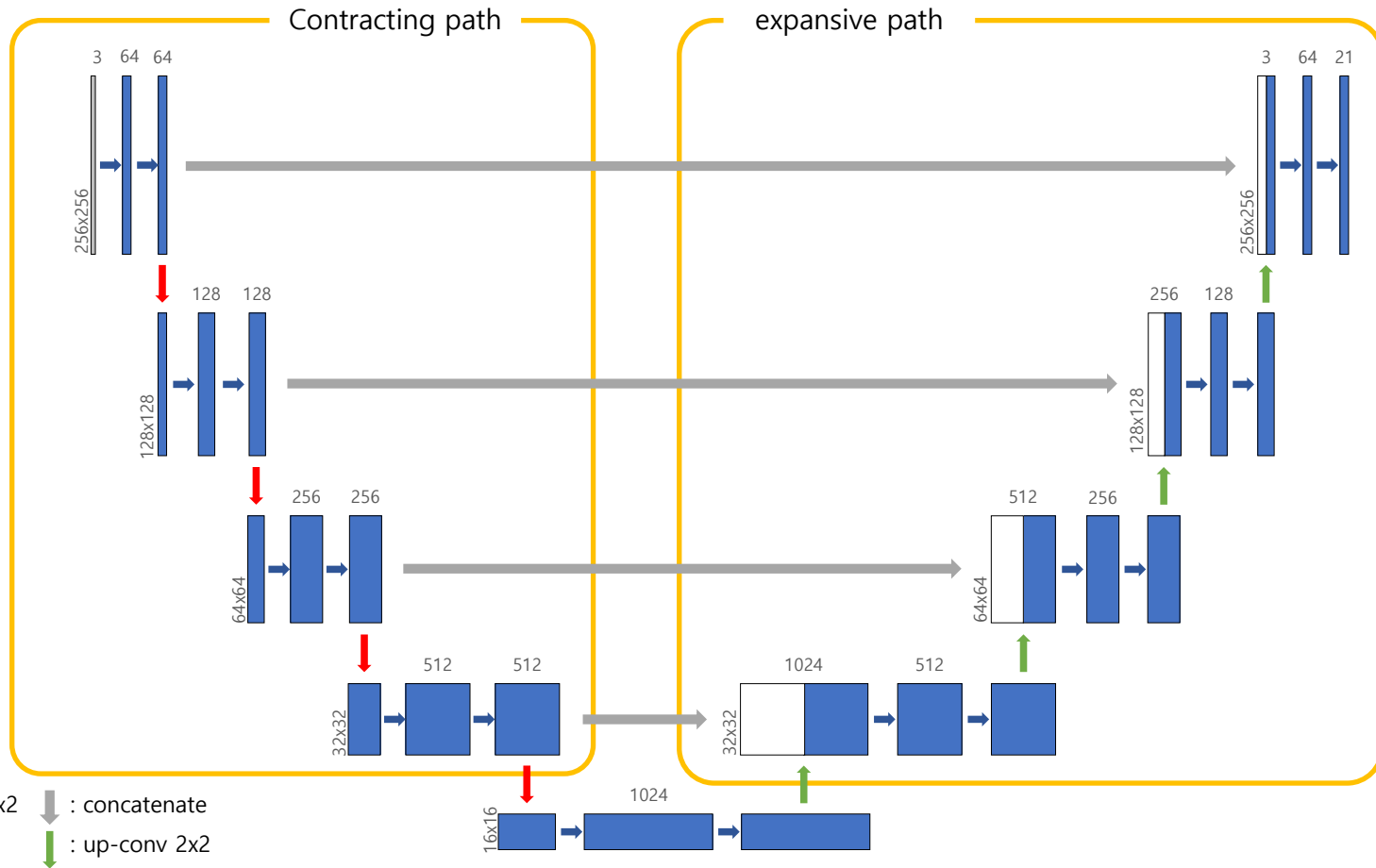
$stride = 16, padding = 8, dilation = 1, kernel\ size = 32, output\ padding = 0$ FCN-16s : 16x16 $\rightarrow$ 256x256

$stride = 8, padding = 4, dilation = 1, kernel\ size = 16, output\ padding = 0$ FCN-8s : 32x32 $\rightarrow$ 256x256

U-Net^[2]

■ Architecture

- Localization을 위해 down sampling의 feature와 upsampling의 feature 결합

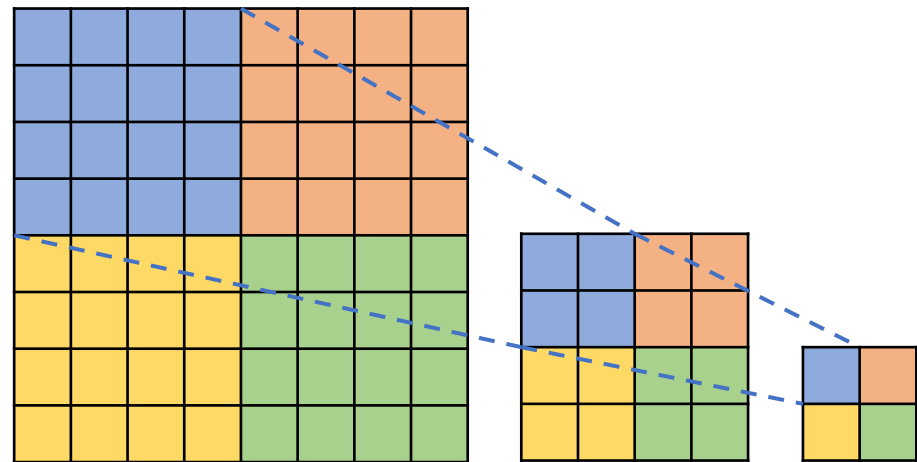
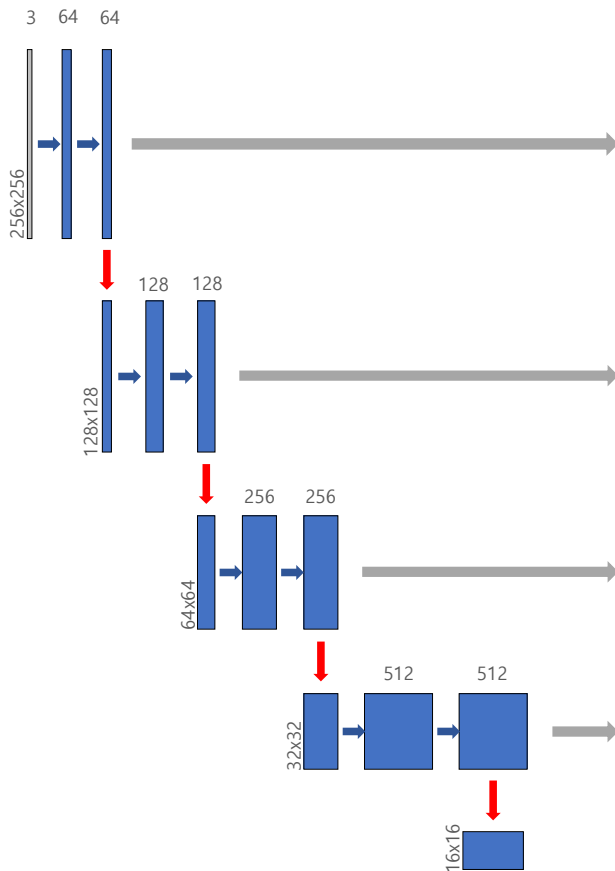


[2] O. Ronneberger, P. Fischer, & T. Brox. "U-net: Convolutional networks for biomedical image segmentation". In *International Conference on Medical image computing and computer-assisted intervention* (pp. 234-241). Cham: Springer international publishing. 2015

U-Net

■ Contracting path

- 입력 이미지로부터 context 정보를 포착하는 역할
- 픽셀마다 입력 이미지의 더 넓은 영역에 대한 정보를 집약



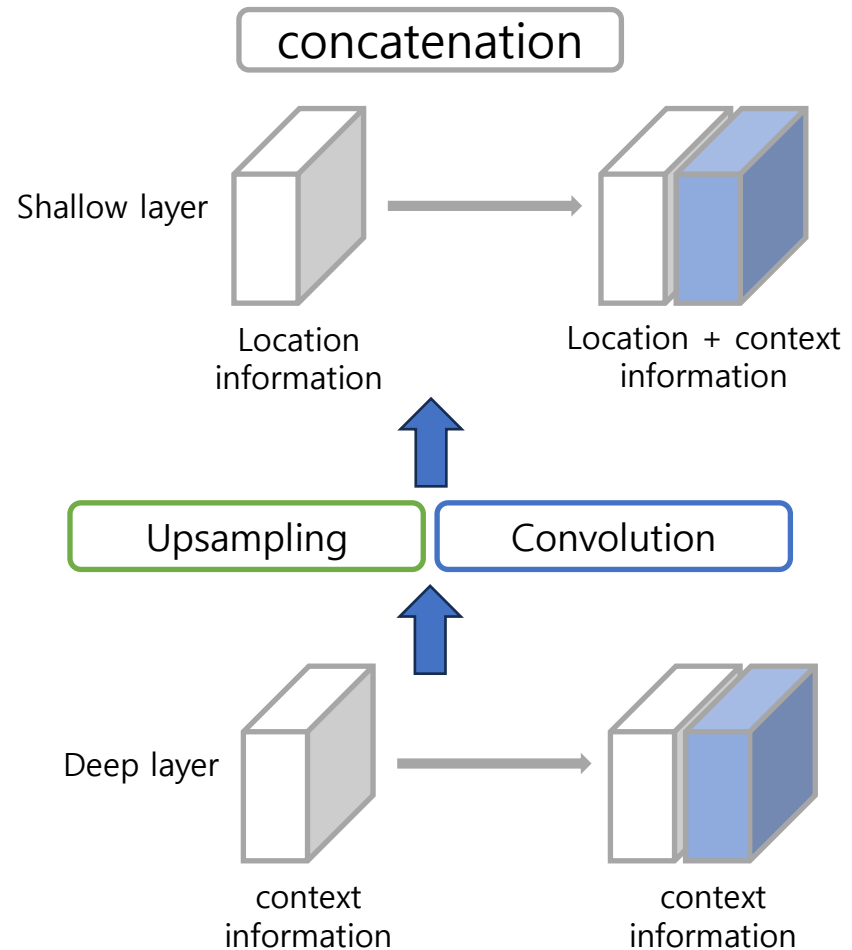
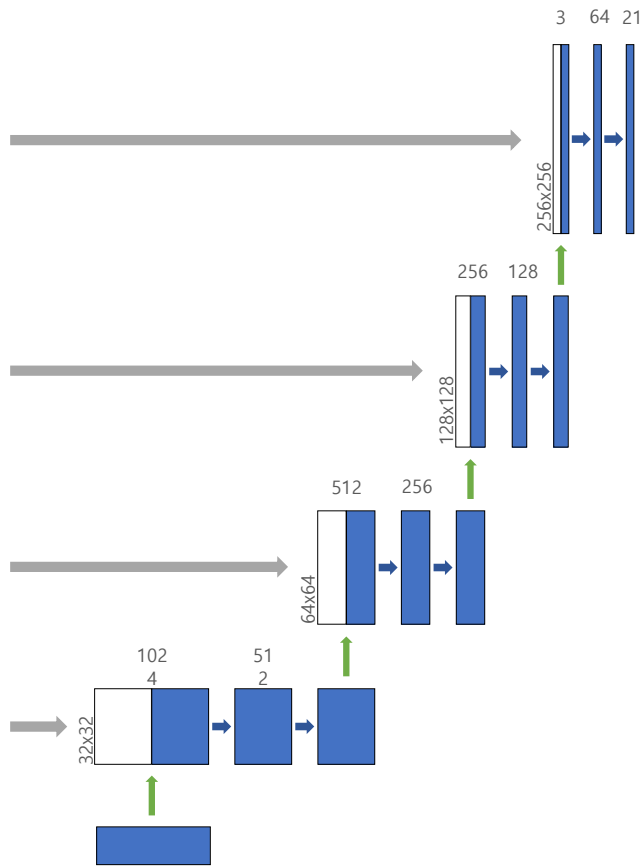
Max pool 2x2

Max pool : capture context information

U-Net

■ Expansive path

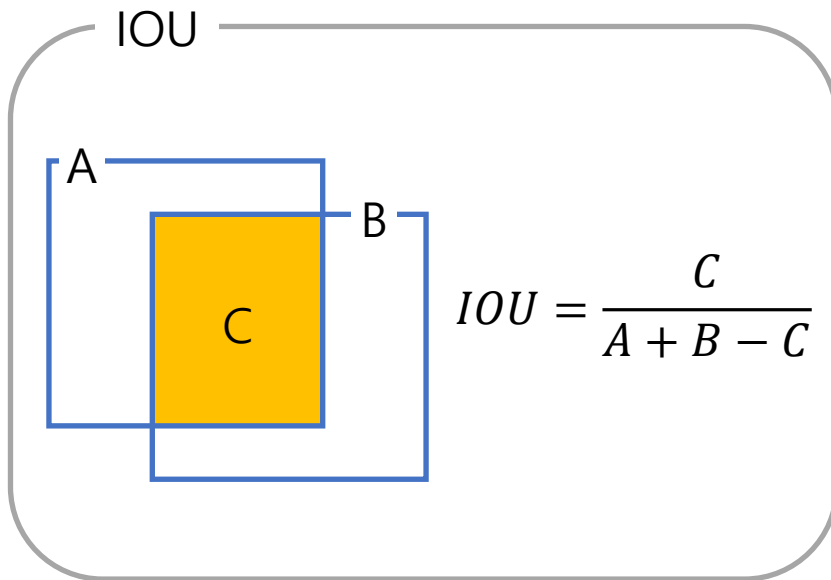
- Coarse feature map으로부터 입력 이미지와 같은 해상도로 복원
- Skip connection으로 context와 location 정보 결합



Experiment

■ Evaluation

- Mean intersection over union(mIOU)



		Predict				
		0	1	2	3	4
Ground Truth	0	5	2	3	0	5
	1	1	2	1	5	1
	2	1	14	7	8	1
	3	1	1	0	11	2
	4	7	2	11	6	2

Class 0

$$IOU = \frac{5}{\sum Predict(0) + \sum G.T(0) - 5}$$



Class 4

mIOU

$$mIOU = \frac{\sum IOU}{number\ of\ class}$$

Experiment

▪ Setting

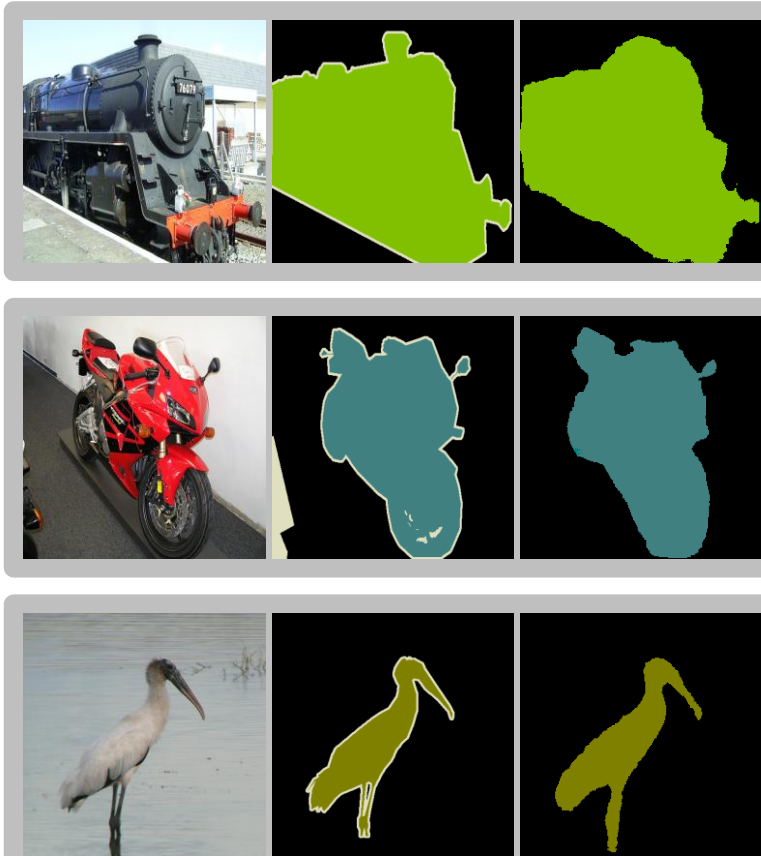
	FCN	U-Net
Dataset	PASCAL VOC2012	PASCAL VOC2012
Epoch	60	60
Batch size	64	32
Learning rate	0.0001	0.00001
Learning rate scheduling	Multiplied by 0.1 every 30 epoch	Multiplied by 0.1 every 30 epoch
Loss function	Cross entropy Loss	Cross entropy Loss
Optimizer	Adam	Adam
Augmentation	Random horizontal Flip Random Crop	Random horizontal Flip Random Crop

Experiment

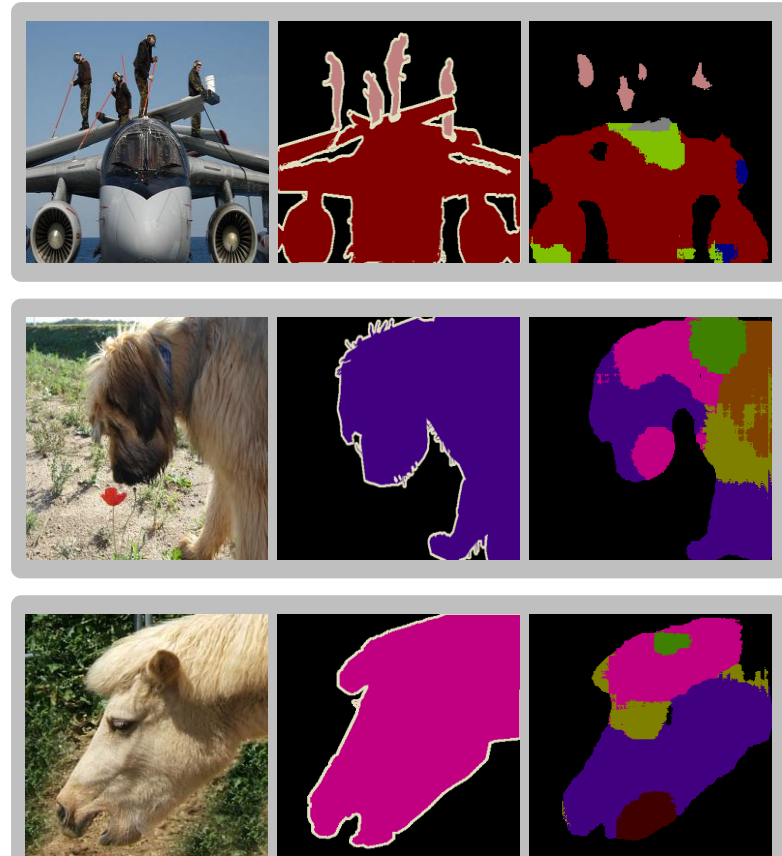
■ FCN-8s Result

	mIOU
FCN-8s	50.25%

Best case



Worst case

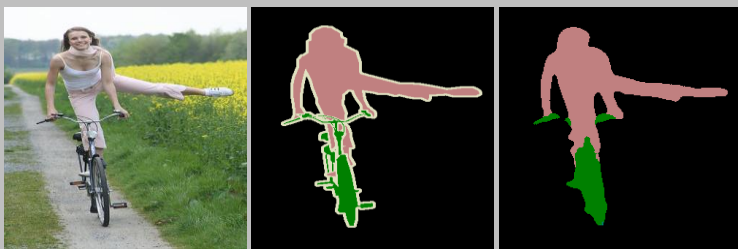
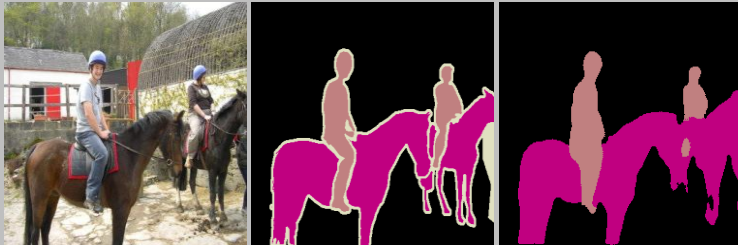


Experiment

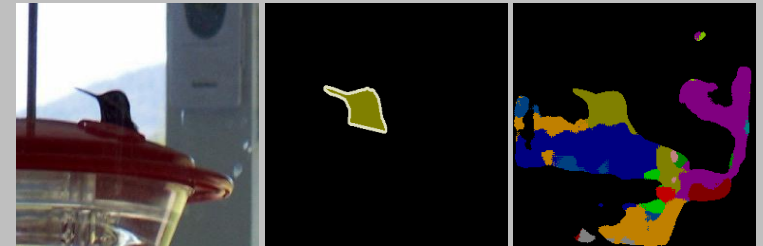
■ U-Net Result

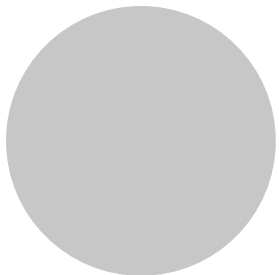
	mIOU
U-Net	59.78%

Best case



Worst case





The end

Thanks

